MORADABAD INSTITUTE OF TECHNOLOGY, MORADABAD
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGG.
Class Test- 1 (First Semester) (Section –A, B, C, D)
(2024-25) (SET-1)

Subject Name: Fundamentals of Electronics Engineering M.M: 20
Subject Code: BEC-101 Duration: 1 hr

Q. No.	1	2	3	4	5	6
CO	1	4	4	4	1	4
Bloom's level	K3	K2	K3	K3	K3	K3

SECTION (A)

1. Define doping. Differentiate between intrinsic and extrinsic semiconductors. (2*1=2)

SECTION (B)

2. Subtract using 1's complement method: $(10111)_2 - (110011)_2$ (3*2=6)
3. What are universal gates and why they are named so? Implement the basic gates with the help of only NAND gate.

SECTION (C)

4. Convert the following numbers into the base indicated (4*3=12)
(a) $(532)_8 = (\quad)_{10}$ (b) $(11011.10)_2 = (\quad)_{10}$
(b) $(101)_{10} = (\quad)_2$ (d) $(11101.011)_2 = (\quad)_{16}$
5. What is depletion layer? Explain the formation of depletion layer under no bias and reverse bias condition with the help of neat labelled diagrams.
6. Simplify the following function using K Map and realize the output with the help of basic gates.
$F(A,B,C,D) = \Sigma\,(0,2,4,5,6,11,15) + \Sigma d\,(1,9,12)$

$3 \times 8^2 + 5 \times 8^1 + 2 \times 8^0$
$300 + 40 + 2$
362

9
10 - A
11 - B
12 - C
13 D

MORADABAD INSTITUTE OF TECHNOLOGY, MORADABAD
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGG.
CT- 2 (First Semester) (Section –A, B, C, D)
(2024-25)

Subject Name: Fundamentals of Electronics Engineering M.M: 20

Subject Code: BEC-101 Duration: 1 hr

Q. No.	1	2	3	4	5	6
CO	2	2	2	2	4	4
Bloom's level	K3	K2	K3	K3	K3	K3

SECTION (A)

Following question carries 2 marks. (2*1=2)

1. Differentiate between Zener breakdown and Avalanche breakdown mechanisms.

SECTION (B)

Each question carries 3 marks. (3*2=6)

2. Write short note on (any two)

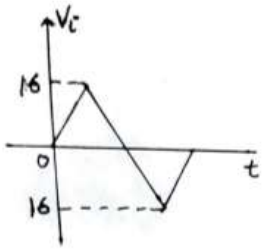
(a) LED (b) Zener diode (c) Photodiode

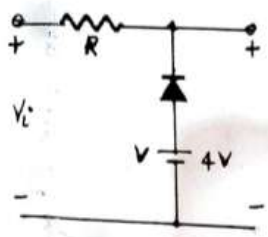
3. Define voltage multiplier circuits. Explain the working of voltage doubler circuit in detail.

SECTION (C)

Each question carries 4 marks. (4*3=12)

4. Determine V_0 for the following network





5. Draw and explain the circuit of Full Wave Center-tap rectifier with the help of necessary waveforms. Also state its advantages and disadvantages.

6. Explain the construction and working of a PNP transistor in CB configuration. Also draw its input and output characteristics.

MORADABAD INSTITUTE OF TECHNOLOGY, MORADABAD
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGG.
Class Test- 3 (1st Semester) (Section –A, B, C, D)
(2024-25) (SET-1)

Subject Name: Fundamentals of Electronics Engineering M.M: 20

Subject Code: BEC-101 Duration: 1 Hr.

Q. No.	1	2	3	4	5	6
CO	3	3	5	2	3	5
Bloom's level	K2	K3	K2	K3	K3	K3

Note:

1. Attempt all questions.
2. Be precise in your answer.

SECTION (A)

Following question carries 2 marks. (2*1=2)

1. Define CMRR. List down the characteristics of an ideal Op-Amp.

SECTION (B)

Each question carries 3 marks. (3*2=6)

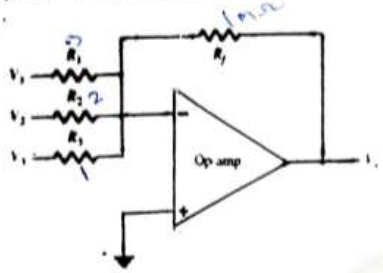
2. Draw and explain the circuit diagram of Op-amp Differentiator.
3. Write Short Notes on

(i) Cellular Communication (ii) Satellite Communication

SECTION (C)

Each question carries 4 marks. (4*3=12)

4. Describe the construction and working of n-channel depletion type MOSFET, with drain and transfer characteristic.
5. Calculate the output voltage of an op-amp summing amplifier for V1 = +3 V, V2 = +2 V, V3 = +1 V, R1 = 3 MΩ, R2 = 2 MΩ, R3 = 1 MΩ & Rf = 1 MΩ.



6. An audio frequency signal 10 sin 2π x 500t is used to amplitude modulate a carrier of 50 sin 2π x 10^5t. Calculate
   (i) Modulation index
   (ii) Sideband frequencies
   (iii) Amplitude of each sideband frequencies
   (iv) Bandwidth required
   (v) Total power delivered to the load of 600Ω
   (vi) Transmission efficiency